Wonjun Lee

Assistant Professor at The Ohio State University

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Research Interests

My research focuses on developing partial differential equations (PDE)-based algorithms to solve high-dimensional machine learning problems and establishing their rigorous theoretical foundations.

Academic Positions

The Ohio State University — *Columbus, OH*
Assistant Professor at The Department of Mathematics — *Aug 2025 - Present*

University of Minnesota, Twin Cities — *Minneaplis, MN*
IMA-NIST Postdoctoral Fellow — *Aug 2022 - Aug 2025*

University of California, Los Angeles — *Los Angeles, CA*
Assistant Adjunct Professor — *Jun 2022 - Aug 2022*

Education

University of California, Los Angeles — *Los Angeles, CA*
Ph.D. in Mathematics Advised by Prof. Stan Osher — *Sep 2017 - Jun 2022*

George Mason University — *Fairfax, VA*
B.S. in Mathematics — *May 2015*

Publications

- **W. Lee**, L. Wang, W. Li, *Generalized Deep JKO: an IMEX particle method for kinetic Fokker-Planck equations*, **Submitted**
- J. Gu, M. Mardani, **W. Lee**, D. Zou, G. Lerman, *Demystifying Latent Space Diffusibility via Fisher Geometry*, **Submitted**
- **W. Lee**, R. O'Neill, D. Zou, J. Calder, G. Lerman, *Geometry-Preserving Encoder/Decoder In Latent Generative Models*, **Submitted**
- J. Calder, **W. Lee**, *Understanding Contrastive Learning through Variational Analysis and Neural Network Optimization Perspectives*, **Submitted**
- **W. Lee**, Y. Yang, D. Zou, G. Lerman, *Monotone Generative Modeling via a Gromov-Monge Embedding*, SIAM Journal on Mathematics of Data Science, 2025
- J. Calder, **W. Lee**, *Monotone Discretizations of Levelset Convex Geometric PDEs*, Numerische Mathematik, 2024
- Y. Yang, **W. Lee**, D. Zou, G. Lerman, *Improving Hyperbolic Representations via Gromov-Wasserstein Regularization*, ECCV, 2024
- **W. Lee**, L. Wang, W. Li, *Deep JKO: Time-Implicit Particle Methods For General Nonlinear Gradient Flows*, Journal of Computational Physics, 2024
- **W. Lee**, S. Liu, W. Li, S. Osher, *Mean Field Control Problems For Vaccine Distribution*, Research in the Mathematical Sciences, 2022
- W. Li, **W. Lee**, S. Osher, *Computation Mean-Field Information Dynamics Associated With Reaction Diffusion Equations*, Journal of Computational Physics, 2022

- S. Agrawal, **W. Lee**, S. W. Fung, L. Nurbekyan, *Random Features for High-Dimensional Nonlocal Mean-Field Games*, Journal of Computational Physics, 2022
- A. Vepa, A. Choi, N. Nakhaei, **W. Lee**, et al. *Weakly-Supervised Convolutional Neural Networks for Vessel Segmentation in Cerebral Angiography*, Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2022
- **W. Lee**, W. Li, B. Lin, A. Monod, *Tropical Optimal Transport and Wasserstein Distances in Phylogenetic Tree Space*, Information Geometry, 2021
- M. Jacobs, **W. Lee**, F. Léger, *The back-and-forth method for Wasserstein gradient flows*, ESAIM: COCV, 27:28, 2021.
- **W. Lee**, S. Liu, H. Tembine, W.C. Li, S. Osher., *Controlling propagation of epidemics via mean-field games*, SIAM Journal on Applied Math, 2020
- **W. Lee**, R.J. Lai, W. Li, S. Osher., *Generalized unnormalized optimal transport and its fast algorithms*, Journal of Computational Physics, 2020
- H. Gao, **W. Lee**, W. Li, Z. Han, S. Osher, and H. V. Poor, *Energy-efficient Velocity Control for Massive Rotary-Wing UAVs: A Mean Field Game Approach*, IEEE Globecom, 2020.
- Y. Kang, S. Liu, **W. Lee**, W. Li, H. Zhang, and Z. Han, *Joint Task Assignment and Trajectory Optimization for a Mobile Robot Swarm by Mean-Field Game*, IEEE Globecom, 2020.

Recent Presentations

2026 Computational Math Seminar, **The Ohio State University**

2026 Control and Optimization Seminar, **Louisiana State University**

2026 2026 AI Research Summit, **The Ohio State University**

2025 Level Set Collective II, **UCLA**

2025 SIAM Central States Section Annual Meeting 2025, **Arkansas, US**

2025 PDE Seminar, **North Carolina State University**

2025 Computational and Applied Mathematics Seminar, **New Jersey Institue of Technology**

2025 Mathematics Colloquium, **Williams College**

2025 Research Seminar, **The Center for Communications Research-La Jolla**

2025 Applied Mathematics Seminar, **Tufts University**

Honors and Awards

2026 NVIDIA Academic Grant: GPU Computing Resources (Co-PI, Jan 2026 – Jun 2026)

2022 Rising Star in Data Science from the University of Chicago. PROFILE LINK.

2021 UCLA Dissertation Year Fellowship ($\$20,000$)

2014 2014 Outstanding Presentation Award at the Joint Mathematical Meetings, Baltimore, MD.

Service and Outreach

2024 Ambassador for the University of Minnesota's Data Science Initiative (DSI), **UMN DSI**

2024 Co-organizer of the minisymposium 'Theory and Applications of Optimal Transport in Machine Learning' with Jeff Calder, **SIAM Annual Meeting 2024**